Erqiang Tang

Ph.D. Candidate, Electronic and Computer Engineering (ECE), HKUST

etangaa@connect.ust.hk

Research Interests

My research lies at the intersection of generative modeling and wireless communications, with a focus on harnessing modern generative AI techniques for wireless signal processing.

Education

The Hong Kong University of Science and Technology (HKUST) 2022 – Present
Ph.D. in Electronic and Computer Engineering
Advisor: Prof. Khaled B. Letaief

The Hong Kong University of Science and Technology (HKUST) 2018 – 2022
B.Eng. in Electronic Engineering, Minor in Design
CGA: 3.98 / 4.30

Publications

* Equal contribution (co-first authors).

1. T. Liao*, **E. Tang***, Z. Wang, P. Biswas, J. Sun, W. Guo, J. Qian, S. Song, J. Zhang, K. B. Letaief. "Scaling Fluid Antenna Systems: Generative Learning and Decentralized Optimization for Efficient Air-Interface Design." *Under review.*

2. **E. Tang**, W. Guo, H. He, S. Song, J. Zhang, K. B. Letaief. "A Unified Two-Stage Generative Diffusion Framework for Channel Estimation and Port Selection in Multiuser MIMO-FAS." *Under review.*

3. **E. Tang**, W. Guo, H. He, S. Song, J. Zhang, K. B. Letaief. "Accurate and Fast Channel Estimation for Fluid Antenna Systems with Diffusion Models." *IEEE Global Communications Conference (GLOBECOM)*, 2025, pp. 2066–2071.

Teaching Experience

Teaching Assistant — Department of Electronic and Computer Engineering, HKUST 2022 – 2023

- EESM 5539 — Wireless Communication Networks
- EESM 5670 — Advanced Architectures and Designs for Communication Networks
- ELEC 3300 — Introduction to Embedded Systems
- ELEC 3100 — Signal Processing and Communications

Delivered tutorial sessions, graded homework and exams, and mentored undergraduate final projects.

Honors & Awards

HKUST Postgraduate Studentship (PGS) 2022 – Present
HKUST Academic Achievement Medal 2022
HKUST School of Engineering Dean's List 2018 – 2021

Professional Service

Journal Reviewer: IEEE Wireless Communications Letters
Conference Reviewer: IEEE Global Communications Conference (GLOBECOM)

Technical Skills

MATLAB, Python (PyTorch), LaTeX

Languages

Mandarin (Native) English (Professional) Cantonese (Conversational)